

HDOC Day-12

Menu-Driven C Program Using switch-case

Problem Statement

Write a menu-driven C program using switch-case to input a positive integer and perform either: (1) find the sum and product of the first and last digits, or (2) count how many digits are lesser than, equal to, and greater than a given digit k.

Algorithm

1. Start.
2. Input a positive integer n.
3. Display the menu: 1. Sum and product of first and last digits; 2. Count digits lesser than, equal to, and greater than k.
4. Input the user choice.
5. If choice = 1: Set last = n % 10. Copy n to temp. Repeatedly divide temp by 10 until temp < 10. Set first = temp. Compute sum = first + last and product = first * last. Display the results.
6. If choice = 2: Input digit k (0 to 9). Copy n to temp and initialize lesser = equal = greater = 0. Repeatedly take digit = temp % 10, compare it with k, increment the appropriate counter, and divide temp by 10. Display all three counts.
7. Otherwise display Invalid choice.
8. Stop.

Test Case Table

How to read the table: Choice 1 tests the first/last digit operation. Choice 2 compares every digit with k. The Expected Result column is what the program should print.

TC	Operation Tested	Number (n)	Choice	Digit k	Expected Result	Why / What is Checked
1	Sum & product of first/last digits	12345	1	N/A	First=1, Last=5 Sum=6, Product=5	Normal multi-digit number
2	Sum & product of first/last digits	7	1	N/A	First=7, Last=7 Sum=14, Product=49	Single-digit number: first and last are same
3	Compare digits with k	12345	2	3	Lesser=2 Equal=1 Greater=2	Digits 1,2 < 3; 3 = 3; 4,5 > 3
4	Compare digits with k	90831	2	8	Lesser=3 Equal=1 Greater=1	Digits 0,3,1 < 8; 8 = 8; 9 > 8
5	Compare digits with k	5555	2	5	Lesser=0 Equal=4 Greater=0	All digits are equal to k
6	Compare digits with k	2468	2	0	Lesser=0 Equal=0	All digits are greater than k

TC	Operation Tested	Number (n)	Choice	Digit k	Expected Result	Why / What is Checked
					Greater=4	

C Program

```
#include <stdio.h>

int main()
{
    int n, choice, temp, digit, k;
    int first, last, sum, product;
    int lesser = 0, equal = 0, greater = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    if (n <= 0)
    {
        printf("Please enter a positive integer.\n");
        return 0;
    }

    printf("\nMENU\n");
    printf("1. Find sum and product of first and last digits\n");
    printf("2. Count digits lesser than, equal to, and greater than k\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            last = n % 10;
            temp = n;
            while (temp >= 10)
                temp = temp / 10;
            first = temp;

            sum = first + last;
            product = first * last;

            printf("First digit = %d\n", first);
            printf("Last digit = %d\n", last);
            printf("Sum = %d\n", sum);
            printf("Product = %d\n", product);
            break;

        case 2:
            printf("Enter digit k (0-9): ");
            scanf("%d", &k);

            if (k < 0 || k > 9)
            {
                printf("k must be a digit from 0 to 9.\n");
                break;
            }

            temp = n;
            while (temp > 0)
            {
                digit = temp % 10;
                if (digit < k)
                    lesser++;
                else if (digit == k)
                    equal++;
            }
        }
    }
}
```

```

        else
            greater++;
        temp = temp / 10;
    }

    printf("Count_lesser = %d\n", lesser);
    printf("Count_equal = %d\n", equal);
    printf("Count_greater = %d\n", greater);
    break;

default:
    printf("Invalid choice.\n");
}

return 0;
}

```

Sample Results

Example 1: Number = 12345, Choice = 1 -> Sum = 6, Product = 5

Example 2: Number = 12345, Choice = 2, k = 3 -> Count_lesser = 2, Count_equal = 1, Count_greater = 2

Screenshots

Screenshot 1: Source Code

```

#include <stdio.h>
int main()
{
    int n, choice, temp, digit, k;
    int first, last, sum, product;
    int lesser = 0, equal = 0, greater = 0;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    if (n <= 0)
    {
        printf("Please enter a positive integer.\n");
        return 0;
    }
    printf("\nMENU\n");
    printf("1. Find sum and product of first and last digits\n");
    printf("2. Count digits lesser than, equal to, and greater than k\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    switch (choice)
    {
        case 1:
            last = n % 10;
            temp = n;
            while (temp >= 10)
            {
                temp = temp / 10;
                first = temp;
                sum = first + last;
                product = first * last;
            }

```

```

            printf("First digit = %d\n", first);
            printf("Last digit = %d\n", last);
            printf("Sum = %d\n", sum);
            printf("Product = %d\n", product);
            break;

```

```

case 2:
printf("Enter digit k (0-9): ");
scanf("%d", &k);
if (k < 0 || k > 9)
{
printf("k must be a digit from 0 to 9.\n");
break;
}
temp = n;
while (temp > 0)
{
digit = temp % 10;
if (digit < k)
lesser++;
else if (digit == k)
equal++;
else
greater++;
temp = temp / 10;
}
printf("Count_lesser = %d\n", lesser);
printf("Count_equal = %d\n", equal);
printf("Count_greater = %d\n", greater);
break;
default:
printf("Invalid choice.\n");
}
return 0;

```

Screenshot 2: Compilation

```

(kali@kali)-[~/Desktop]
$ vi sample4.c

(kali@kali)-[~/Desktop]
$ gcc sample4.c -o sample4 -lm

```

Screenshot 3: Execution - Choice 1

```

(kali@kali)-[~/Desktop]
$ ./sample4
Enter a positive integer: 12345

MENU
1. Find sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k
Enter your choice: 1
First digit = 1
Last digit = 5
Sum = 6
Product = 5

```

```

(kali@kali)-[~/Desktop]
$ ./sample4
Enter a positive integer: 7

MENU
1. Find sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k
Enter your choice: 1
First digit = 7
Last digit = 7
Sum = 14
Product = 49

```

Screenshot 4: Execution - Choice 2

```
(kali㉿kali)-[~/Desktop]
$ ./sample4
Enter a positive integer: 12345

MENU
1. Find sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k
Enter your choice: 2
Enter digit k (0-9): 3
Count_lesser = 2
Count_equal = 1
Count_greater = 2
```

```
(kali㉿kali)-[~/Desktop]
$ ./sample4
Enter a positive integer: 90831

MENU
1. Find sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k
Enter your choice: 2
Enter digit k (0-9): 8
Count_lesser = 3
Count_equal = 1
Count_greater = 1
```

```
(kali㉿kali)-[~/Desktop]
$ ./sample4
Enter a positive integer: 5555

MENU
1. Find sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k
Enter your choice: 2
Enter digit k (0-9): 5
Count_lesser = 0
Count_equal = 4
Count_greater = 0
```

```
(kali㉿kali)-[~/Desktop]
$ ./sample4
Enter a positive integer: 2468

MENU
1. Find sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k
Enter your choice: 2
Enter digit k (0-9): 0
Count_lesser = 0
Count_equal = 0
Count_greater = 4
```

Conclusion

The menu-driven program successfully uses switch-case to perform the selected digit operation and was tested against the prepared test cases.